



Moon and Venus

The planets and the Moon all move within the zodiac band of constellations and so can often be viewed together. Here, the Moon and Venus are seen in the early evening sky.

Your view of the solar system

The nearest celestial objects to Earth are members of the solar system—a group of eight planets and smaller objects that orbit the Sun. The planets and our Moon emit no light of their own but shine by reflecting sunlight. From Earth we can see five of these planets with relative ease. As the planets, Moon, and Earth travel around the Sun, our view changes, at times creating spectacular sights for us on Earth.

VIEWING THE PLANETS

Of the eight planets, Earth is the third in distance from the Sun. Mercury and Venus are closer and, for this reason, are called the inferior planets. The five other planets—Mars, Jupiter, Saturn, Uranus, and Neptune—are all farther away than Earth and are called the superior planets. The five visible to the naked eye are Mercury, Venus, Mars, Jupiter, and Saturn, and their existence has been known since humans first looked into space. Uranus's brightness suggests that it is just within the range of

naked-eye visibility, but it is difficult to see. Uranus was only discovered in 1781, and then through a telescope. Neptune, farthest from the Sun, is too faint to see with the naked eye.

All eight planets follow paths around the Sun, and one complete circuit of the Sun is called an orbit. As each planet orbits, and time passes, the positions of the planets relative to Earth change. The view that we have from Earth of a particular planet depends on these relative positions. For instance, when a planet is on the opposite side of the Sun from Earth, it is lost from view. The planet is still present in the daytime sky, but it is invisible because it is drowned out by the glare of the Sun. For this reason, Mercury and Venus are also invisible when situated between the Sun and Earth. Specific juxtapositions of the Earth, the Sun, and the planets have names. These act as a quick indicator of whether or not a planet is visible. They can also signify the phase, brightness, size, and time of visibility of a planet.

The close orbits of Mercury and Venus to the Sun mean that these two planets are never far from the Sun in the sky. They are best seen when at locations that are known as greatest elongation. At this point, they are at their maximum angle to the east or west of the Sun. The superior planets are best seen at opposition, when they are on the opposite side of the Earth from the Sun. In this position, a planet is close to Earth and appears particularly large and bright. At opposition, a planet is visible all night long. It is found to the south at midnight by observers in the northern hemisphere, and to the north at midnight by observers in the southern hemisphere.

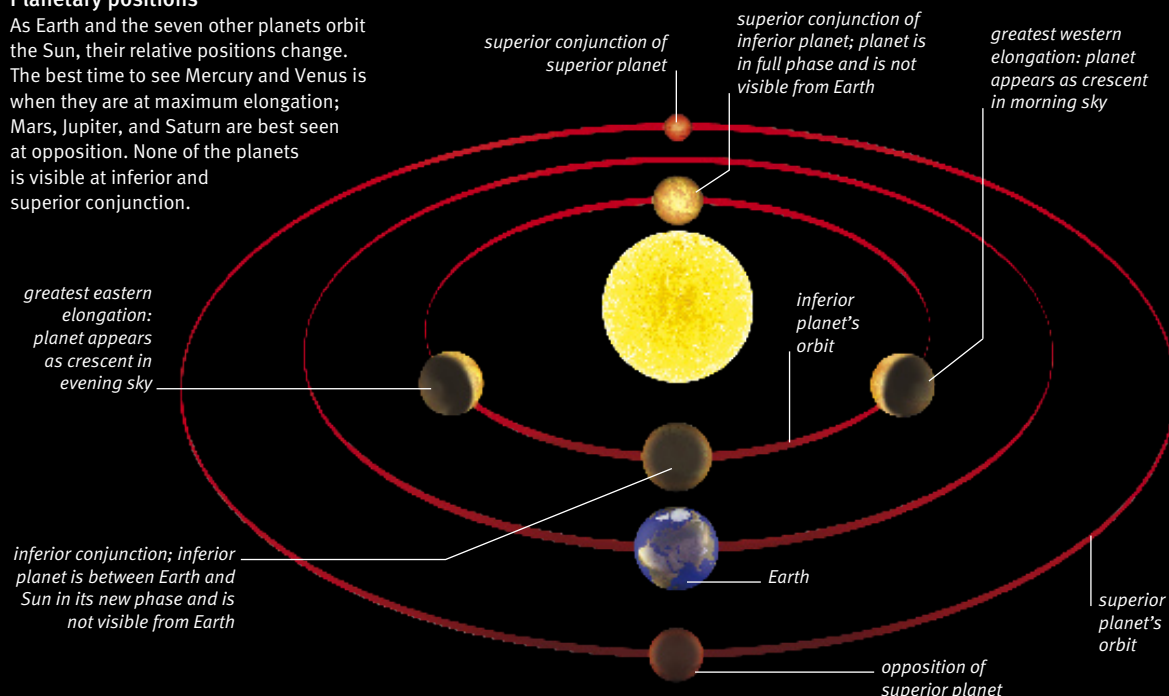
RETROGRADE MOTION

The planets make their stately progress against the starry backdrop from west to east. However, planets are occasionally seen to move in the opposite direction, from east to west, with their paths seeming to loop or zigzag across the sky. This backward movement, known as retrograde motion, is nothing more than an illusion created by our viewpoint on Earth. It is achieved when the faster-moving planet Earth

overtakes a slower-moving superior planet, such as Mars. As Earth moves along its orbit, the superior planet is seen from a different perspective. Superior planets go into retrograde motion around the time of opposition.

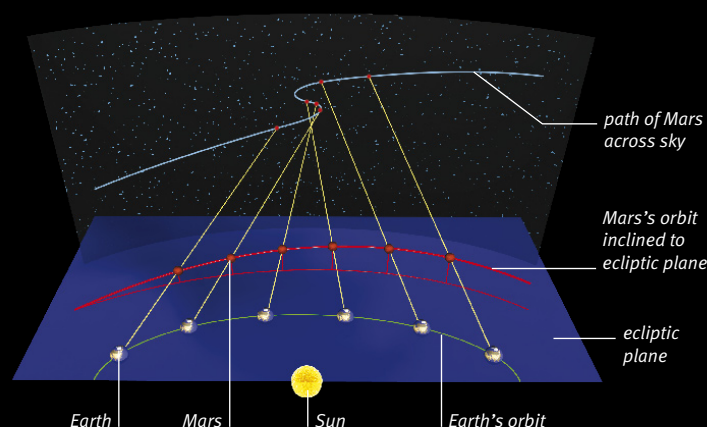
Planetary positions

As Earth and the seven other planets orbit the Sun, their relative positions change. The best time to see Mercury and Venus is when they are at maximum elongation; Mars, Jupiter, and Saturn are best seen at opposition. None of the planets is visible at inferior and superior conjunction.



Mars looping the loop

The retrograde, or backward, movement of a planet across Earth's sky is an illusion caused by our viewpoint from Earth. It is achieved when the faster-moving planet Earth overtakes a superior planet—in this case, Mars, as each moves from west to east (right to left in this diagram). As Earth overtakes on the inside, Mars appears to perform a zigzag across the sky.



MARS SEEN OVER SEVERAL MONTHS